Data Mart Implementation (P01)

DECISION SUPPORT SYSTEMS, 2025-26

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1. Introduction

The goal of this project is to design and implement a **data mart** based on the operational database of *Wide World Importers* (WWI), a fictitious wholesale novelty-goods importer headquartered in San Francisco, US. WWI's customers are mostly companies who resell to individuals — specialty stores, supermarkets, computing stores, tourist-attraction shops — across the United States.

The OLTP source covers WWI's full sales workflow: customers place **orders**, those orders are converted into **invoices** (with **invoice lines** at item granularity), items are physically moved through one of several **delivery methods**, and customers settle invoices via **customer transactions** linked to **payment methods**. When stock runs out, items are **backordered** and shipped later in a separate shipment. This work focuses on the *post-order* portion of the workflow — invoicing, fulfilment and the goods sold — so the data mart can answer questions such as *"Which products topped sales last month?"*, *"Which sales territories grew the most?"* or *"How does invoice volume vary by delivery method and salesperson?"*.

The deliverables include: (i) the dimensional model (Data Warehouse matrix, ER diagram, data description maps), (ii) the implementation of the **ELT process under the medallion architecture** (Bronze → Silver → Gold) on a Postgres target, and (iii) automated **data-quality checks** that validate the loaded mart.

2. Data sources

The operational source is the **WWI sample database**, hosted on PostgreSQL at `postgres2.ipca.pt` (database `wwi`). The relational model is the *adapted* WWI ER diagram presented in the project guide: it covers `customers`, `customercategories`, `buyinggroups`, `people` (employees and contacts), `cities` / `stateprovinces` / `countries` (geography), `stockitems` / `stockgroups` / `colors` / `packagetypes` (catalog), `orders` / `orderlines` (orders), `invoices` / `invoicelines` (billing), `customertransactions` / `paymentmethods` / `transactiontypes` (financial events) and `deliverymethods` (logistics).

Data profiling was performed against the live source with `scripts/data_profiling.py` (VPN ON) and re-validated against the offline snapshots stored in `tmp_snapshots/`. The relevant business objects and their volumes are summarized below.

Event / object	Table	Nr. records	Nr. cols	PK	PK uniqueness
Buying groups	buyinggroups	2	2	buyinggroupid	100.00%
Cities	cities	37,940	5	cityid	100.00%
Colors	colors	36	2	colorid	100.00%
Countries	countries	190	8	countryid	100.00%
Customer categories	customercategories	8	2	customercategoryid	100.00%
Customers	customers	663	28	customerid	100.00%
Customer transactions	customertransactions	97,147	12	customertransactionid	100.00%
Delivery methods	deliverymethods	10	2	deliverymethodid	100.00%
Invoice lines	invoicelines	228,265	11	invoicelineid	100.00%
Customer invoices	invoices	70,510	23	invoiceid	100.00%
Order lines	orderlines	231,412	9	orderlineid	100.00%
Customer orders	orders	73,595	14	orderid	100.00%
Package types	packagetypes	14	2	packagetypeid	100.00%

Event / object	Table	Nr. records	Nr. cols	PK	PK uniqueness
Payment methods	paymentmethods	4	2	paymentmethodid	100.00%
Employees and contacts	people	1,111	18	personid	100.00%
Special deals	specialdeals	2	12	specialdealid	100.00%
State provinces	stateprovinces	53	6	stateprovinceid	100.00%
Stock groups	stockgroups	10	2	stockgroupid	100.00%
Products / stock items	stockitems	227	22	stockitemid	100.00%
Transaction types	transactiontypes	13	2	transactiontypeid	100.00%

Table 1: Summary of WWI database contents

Profiling findings (relevant to the mart):

- All primary keys present 100% uniqueness, so business keys can be reused as join columns into the dimensional layer.
- `paymentmethodid` lives on `customertransactions` (not on `invoices`), so payment information is collected from the transactions side, not directly from invoice headers.
- The `countries` snapshot exhibits a column-name typo (`ountryname`); this is corrected during the silver transform.
- `invoicelines` (228,265 rows) and `invoices` (70,510 rows) are the highest-volume operational events, so they drive the grain of the two fact tables.

3. Dimensional modelling

Goals and analytical questions

The data mart was designed to support a self-service analytic layer answering questions such as:

- Which **stock items** generated the highest revenue last month / quarter / year?
- Which **customer categories** or **sales territories** show the largest sales growth?
- Which **salespersons** are top performers (by invoice volume and by line value)?
- How does the choice of **delivery method** affect invoice value and on-time fulfilment?
- What is the **invoice amount** distributed by **customer**, **date**, **location** and **delivery method**?
- What is the breakdown of **payments** by **payment method** and **transaction type** (auxiliary, served from `customertransactions`)?

These questions translate to two confirmed business processes — **Sales** (line-level) and **Invoicing** (header-level) — sharing conformed dimensions for time, customer, employee and location.

Data Warehouse Matrix

BUSINESS PROCESS \ DIMENSION	DimDate	DimCustomer	DimEmployee	DimProduct	DimLocation	DimDeliveryMethod	DimPaymentMethod
Sales (FactSales) — invoice line	X	X	X	X	X		
Invoicing (FactInvoices) — invoice header	X	X	X		X	X	
Payments (auxiliary, customertransactions)	X	X					X

Table 2: Data Warehouse Matrix

The two primary fact tables loaded into Gold are `FactSales` and `FactInvoices`. The third row (Payments) is currently materialized at the silver level via `silver.stg_*` for completeness; it can be promoted to Gold as `FactPayments` if the analytical scope is later widened.

4. Design of the dimensional data model

Fact tables — granularity and measures

FactSales — *grain: one row per invoice line* (one row per `invoicelines.invoicelineid`).

Measure	Type	Derivation
Quantity	Additive	Direct from invoicelines.quantity
UnitPrice	Non-additive (avg)	Direct from invoicelines.unitprice
TaxAmount	Additive	Direct from invoicelines.taxamount
ExtendedPrice	Additive	Direct from invoicelines.extendedprice (= quantity × unitprice + tax)
LineProfit	Additive	Direct from invoicelines.lineprofit

FactInvoices — *grain: one row per invoice header* (one row per `invoices.invoiceid`).

Measure	Type	Derivation
InvoiceAmount	Additive	SUM(invoicelines.extendedprice) over the invoice
PaymentDelay_Days	Semi-additive	MIN(customertransactions.transactiondate) – invoices.invoicedate
OutstandingBalance	Semi-additive	InvoiceAmount – SUM(customertransactions.transactionamount)

`PaymentDelay_Days` and `OutstandingBalance` are **derived measures** computed during the silver-to-gold step by joining `customertransactions` against `invoices`.

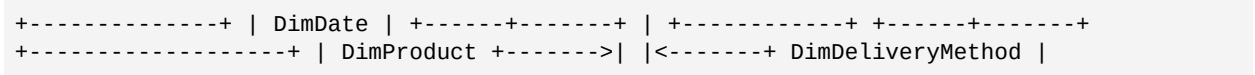
Dimensions and SCD policy

Dimension	SCD type	Justification
DimDate	Static (Type 0)	Calendar dimension generated programmatically; no history needed.
DimCustomer	Type 2	Customers can change category, name, etc.; history matters for trend analysis.
DimEmployee	Type 2	Salesperson role and full name change over time.
DimProduct	Type 2	Stock items can be renamed / rebranded.
DimLocation	Type 2	Composite of cities × stateprovinces × countries; treated as Type 2 for consistency.
DimDeliveryMethod	Type 2 (small)	Track renames; only 10 rows.
DimPaymentMethod	Type 2 (small)	Track renames; only 4 rows.

Surrogate keys (auto-incrementing `SERIAL`) decouple the dimensional model from the OLTP business keys and enable Type 2 history (multiple rows per business key, controlled by `version` / `date_from` / `date_to`).

ER Diagram (star schema)

The star schema is composed of two fact tables (**FactSales**, **FactInvoices**) and seven dimensions sharing conformed surrogate keys. The diagram is reproduced as a textual map below; the Mermaid source is bundled in `REPORT.md` for graphical rendering.



```
+------------+ | | +-------------------+ | FactSales | +-------------+ | | | DimCustomer
+------>| | +-------------+ | | +------------------+ | | | FactInvoices |<--+ DimEmployee
+-------------+ | | | | +-----------+ | DimEmployee +------>+------+-------+ | |
+-------------+ ^ +---------+--------+ | ^ +--------------+ | | | DimLocation
+-------------+-------------------------+ +--------------+ (DimPaymentMethod is populated
from customertransactions; it is conformed but joined indirectly to FactInvoices via
invoiceid mapping.)
```

Data description maps for the two reference dimensions (DimCustomer and DimProduct) are presented in **Appendix A**.

5. Data mart implementation

The pipeline follows the **medallion architecture** (Bronze → Silver → Gold) on a Postgres DWH (Supabase). Source extraction (over VPN, against `postgres2.ipca.pt`) and DWH loading (off VPN) are split across separate notebook cells so the operator toggles VPN exactly once.

```
public.* (WWI source — VPN ON) | save -> CSV + schema JSON in tmp_snapshots/,
tmp_increments/ v bronze.* (full-fidelity copy — VPN OFF) | read active snapshot, clean,
conform v silver.stg_* (staging, project-only columns — VPN OFF) | SCD Type 2 for dims,
surrogate-key lookups for facts v gold.* (star schema — VPN OFF)
```

Notebooks and transformations

Notebook	Layer	Responsibility
00_setup.ipynb	meta	Loads .env, builds engines, creates bronze/silver/gold schemas, deploys bronze._load_control and the gold DDL.
01_bronze.ipynb	bronze	Two-phase pattern. Extract (VPN ON): dumps each source table to tmp_snapshots/.csv with schema JSON. Apply (VPN OFF): rebuilds bronze tables from the JSON, loads CSVs and registers each load into bronze._load_control.
02_silver.ipynb	silver	DROP+CREATE staging tables, applies clean_str to free-text fields, fixes the ountryname → country typo, joins customercategories, joins cities × stateprovinces × countries, computes derived measures.
03_gold.ipynb	gold	Generates DimDate, calls load_scd2() per dimension (uses row_hash() to detect changes), then loads FactSales and FactInvoices via surrogate-key lookup against the current dim row (date_to IS NULL). Indexes/constraints applied from scripts/create_indexes_and_constraints.sql.

Key ELT functions

- `make_engine()` — factory returning the right SQLAlchemy engine for source vs DWH.
- `register_load()` — writes one row to `bronze._load_control` per applied snapshot/increment.
- `compute_row_hash()` / `row_hash()` — canonicalize values and hash with MD5 over '|'-joined SCD2 attributes.
- `load_scd2()` — closes the previous current row (`date_to = run_at`) and inserts `version + 1` when a hash change is detected; otherwise no-ops.

Quality checks (04_quality_checks.py)

For every load, the script writes `quality_report.json` and exits non-zero if any check fails:

- **Orphan FKs** in both fact tables.
- **Unmapped FKs** — silver business keys with no current dim row (silent data-loss detection).
- **Duplicate current dim rows** (must be 0 thanks to partial-unique indexes).
- **Null percentages** on business keys vs `THRESHOLD_NULL_PERCENT = 5.0`.
- **Row-count reconciliation** between `silver.stg_*` and `gold.dim* (current)`.

Data mart content — summary of loaded rows (run: 2026-05-05)

Gold table	Current rows	Notes
gold.DimDate	(calendar)	Snapshot dimension covering invoice / transaction date range.

Gold table	Current rows	Notes
gold.DimEmployee	1,111	SCD2; reconciled silver→gold 1,111 ↔ 1,111.
gold.DimCustomer	663	SCD2; reconciled 663 ↔ 663.
gold.DimProduct	227	SCD2; reconciled 227 ↔ 227.
gold.DimLocation	37,940	SCD2; composite city × state × country, reconciled 37,940 ↔ 37,940.
gold.DimDeliveryMethod	10	Reconciled 10 ↔ 10.
gold.DimPaymentMethod	4	Reconciled 4 ↔ 4.
gold.FactSales	228,265	Grain = invoice line, derived from invoicelines.
gold.FactInvoices	70,510	Grain = invoice header, derived from invoices.

Quality verdict: all orphan-FK counts = 0, all unmapped-FK counts = 0, no duplicate current rows and 0% nulls on every business key. The mart is internally consistent.

6. Conclusion

Strengths.

- End-to-end medallion architecture (Bronze → Silver → Gold) with clean separation between source extraction (VPN ON) and DWH loading (VPN OFF).
- Seven dimensions and two facts following textbook Kimball patterns: SERIAL surrogate keys, SCD Type 2 with version / date_from / date_to, partial-unique indexes guarding the current row.
- **Idempotent pipeline**: silver tables are DROP+CREATE on every run, gold dimensions merged via SCD2 hashing, bronze._load_control records every apply for full traceability.
- Automated quality checks (04_quality_checks.py + 99_verification.py) close the loop: a failing FK or duplicate dim row breaks the build, not the report.

Weaknesses / known limitations.

- DimDate is loaded as a snapshot covering only the observed invoice date range; a fully-attributed calendar (fiscal year, holidays, weekday names) would broaden time-based analysis.
- Payments are exposed only via silver staging and the auxiliary customertransactions mapping; a dedicated FactPayments would make payment-method analytics first-class.
- Source-side schema typos (ountryname) are silently fixed in silver but not pushed back as a data-quality issue to the source owners.
- OrderLines and the fulfilment side (backorders) are not modelled; the mart focuses on invoiced sales only.

Future work.

- Promote silver.stg_fact_invoices payment join into a Gold FactPayments table.
- Extend DimDate with a generated 2010–2030 range and standard fiscal/holiday attributes.
- Add aggregate / cumulative tables (agg_sales_monthly, agg_sales_by_territory) for dashboard latency.
- Wire the quality report into a CI step so an ERROR row in bronze._load_control blocks downstream layers automatically.

7. Bibliography

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Appendix A — Data description maps

This appendix documents the column-level mapping between the source (OLTP) and the data mart (Gold) for the two reference dimensions, **DimCustomer** and **DimProduct**.

Table 3 — Data description map of DimCustomer

Name	Type of table	Nr. records	Description
DimCustomer	Dimension	663 (current)	Customers of WWI, with category resolved from customercategories. SCD Type 2: history preserved via version / date_from / date_to.

Column	Description	Data type	SCD	Src table	Src column	Src type	ETL rules	Example
CustomerKey	Surrogate key	INT (SERIAL)	1	—	—	—	Auto-generated by Postgres SERIAL.	1, 2, 3, …
CustomerID	Business key	INT	1	customers	customerid	int4	Direct copy.	832
CustomerName	Customer name	VARCHAR(255)	2	customers	customername	varchar	clean_str (trim, empty → NULL).	Alvin Bollinger
Category	Customer category	VARCHAR(255)	2	customercategories	customercategoryname	varchar	LEFT JOIN on customercategoryid, then clean_str.	Novelty Shop
version	SCD2 version	INT	—	—	—	—	Set to 1 on insert; +1 on hash diff.	1, 2
date_from	Validity start	TIMESTAMP	—	—	—	—	run_at of the load that introduced the row.	2026-05-04 22:00:00
date_to	Validity end	TIMESTAMP	—	—	—	—	NULL = current; set on UPDATE detection.	NULL

SCD Type 2 detection. Each silver row is hashed over [CustomerName, Category]; if the hash differs from the current gold row for the same CustomerID, the gold row is closed (date_to = run_at) and a new row is inserted with version + 1.

Table 4 — Data description map of DimProduct

Name	Type of table	Nr. records	Description
DimProduct	Dimension	227 (current)	Stock items sold by WWI, including commercial brand. SCD Type 2 to capture renames / rebrandings.

Column	Description	Data type	SCD	Src table	Src column	Src type	ETL rules	Example
ProductKey	Surrogate key	INT (SERIAL)	1	—	—	—	Auto-generated by Postgres SERIAL.	1, 2, 3, …
StockItemID	Business key	INT	1	stockitems	stockitemid	int4	Direct copy.	11
StockItemName	Item name	VARCHAR(255)	2	stockitems	stockitemname	varchar	clean_str (trim, empty → NULL).	Spy uniform
Brand	Brand	VARCHAR(255)	2	stockitems	brand	varchar	clean_str; may be NULL for unbranded.	Northwind, NULL
version	SCD2 version	INT	—	—	—	—	Same as DimCustomer.	1, 2
date_from	Validity start	TIMESTAMP	—	—	—	—	run_at of the load.	2026-05-04 22:00:00

Column	Description	Data type	SCD	Src table	Src column	Src type	ETL rules	Example
date_to	Validity end	TIMESTAMP	—	—	—	—	NULL for current row.	NULL

SCD legend: 1 = SCD Type 1 (overwrite); 2 = SCD Type 2 (track history); — = technical column.

Appendix B — SQL DW Script

Full DDL — schemas, bronze._load_control, the seven dimensions, both facts, indexes and partial-unique constraints. Maintained at `scripts/dw_script.sql`.

```
-- ================================================================================
-- DSS P01 – Wide World Importers Data Mart
-- Consolidated DW SQL Script (Appendix B of the report)
-- ================================================================================
-- Section 1: Schemas (medallion architecture)
-- Section 2: Bronze metadata table (_load_control)
-- Section 3: Gold dimensions (SCD2 + DimDate)
-- Section 4: Gold fact tables
-- Section 5: Indexes and partial-unique constraints
-- ================================================================================

-- --------------------------------------------------------------------------------
-- Section 1 – Schemas
-- --------------------------------------------------------------------------------
CREATE SCHEMA IF NOT EXISTS bronze;
CREATE SCHEMA IF NOT EXISTS silver;
CREATE SCHEMA IF NOT EXISTS gold;

-- --------------------------------------------------------------------------------
-- Section 2 – Bronze metadata: load control
-- --------------------------------------------------------------------------------
CREATE TABLE IF NOT EXISTS bronze._load_control (
table_name VARCHAR(100) NOT NULL,
strategy VARCHAR(20) NOT NULL,
snapshot_id INT,
watermark_date DATE,
loaded_at TIMESTAMP NOT NULL,
rows_total INT,
rows_inserted INT,
rows_updated INT,
rows_deleted INT,
status VARCHAR(20) NOT NULL,
PRIMARY KEY (table_name, loaded_at)
);

-- --------------------------------------------------------------------------------
-- Section 3 – Gold dimensions
-- --------------------------------------------------------------------------------

-- SCD Type 2: DimEmployee
CREATE TABLE IF NOT EXISTS gold.DimEmployee (
EmployeeKey SERIAL PRIMARY KEY,
version INT NOT NULL,
date_from TIMESTAMP NOT NULL,
date_to TIMESTAMP,
PersonID INT NOT NULL,
FullName VARCHAR(255),
IsSalesperson INT
);

-- SCD Type 2: DimCustomer
CREATE TABLE IF NOT EXISTS gold.DimCustomer (
CustomerKey SERIAL PRIMARY KEY,
version INT NOT NULL,
date_from TIMESTAMP NOT NULL,
date_to TIMESTAMP,
CustomerID INT NOT NULL,
CustomerName VARCHAR(255),
Category VARCHAR(255)
);
```

```
-- SCD Type 2: DimLocation
CREATE TABLE IF NOT EXISTS gold.DimLocation (
LocationKey SERIAL PRIMARY KEY,
version INT NOT NULL,
date_from TIMESTAMP NOT NULL,
date_to TIMESTAMP,
LocationID INT NOT NULL,
City VARCHAR(255),
State VARCHAR(255),
Country VARCHAR(255),
SalesTerritory VARCHAR(50)
);

-- DimDate (snapshot, no SCD)
CREATE TABLE IF NOT EXISTS gold.DimDate (
DateKey INT PRIMARY KEY,
Date DATE,
Year INT,
Month INT,
Day INT
);

-- SCD Type 2: DimProduct
CREATE TABLE IF NOT EXISTS gold.DimProduct (
ProductKey SERIAL PRIMARY KEY,
version INT NOT NULL,
date_from TIMESTAMP NOT NULL,
date_to TIMESTAMP,
StockItemID INT NOT NULL,
StockItemName VARCHAR(255),
Brand VARCHAR(255)
);

-- SCD Type 2: DimDeliveryMethod
CREATE TABLE IF NOT EXISTS gold.DimDeliveryMethod (
DeliveryMethodKey SERIAL PRIMARY KEY,
version INT NOT NULL,
date_from TIMESTAMP NOT NULL,
date_to TIMESTAMP,
DeliveryMethodID INT NOT NULL,
DeliveryMethodName VARCHAR(255)
);

-- SCD Type 2: DimPaymentMethod
CREATE TABLE IF NOT EXISTS gold.DimPaymentMethod (
PaymentMethodKey SERIAL PRIMARY KEY,
version INT NOT NULL,
date_from TIMESTAMP NOT NULL,
date_to TIMESTAMP,
PaymentMethodID INT NOT NULL,
PaymentMethodName VARCHAR(255)
);

-- ------------------------------------------------------------------------
-- Section 4 — Gold fact tables
-- ------------------------------------------------------------------------

-- FactSales — grain: invoice line
CREATE TABLE IF NOT EXISTS gold.FactSales (
SalesKey SERIAL PRIMARY KEY,
InvoiceID INT,
DateKey INT,
EmployeeKey INT,
CustomerKey INT,
ProductKey INT,
LocationKey INT,
Quantity INT,
```

```
UnitPrice NUMERIC(10,2),
TaxAmount NUMERIC(10,2),
ExtendedPrice NUMERIC(10,2),
LineProfit NUMERIC(10,2),
FOREIGN KEY (DateKey) REFERENCES gold.DimDate(DateKey),
FOREIGN KEY (EmployeeKey) REFERENCES gold.DimEmployee(EmployeeKey),
FOREIGN KEY (CustomerKey) REFERENCES gold.DimCustomer(CustomerKey),
FOREIGN KEY (ProductKey) REFERENCES gold.DimProduct(ProductKey),
FOREIGN KEY (LocationKey) REFERENCES gold.DimLocation(LocationKey)
);

-- FactInvoices — grain: invoice header
CREATE TABLE IF NOT EXISTS gold.FactInvoices (
InvoiceFactKey SERIAL PRIMARY KEY,
InvoiceID INT,
DateKey INT,
EmployeeKey INT,
CustomerKey INT,
LocationKey INT,
AccountsEmployeeKey INT,
DeliveryMethodKey INT,
InvoiceAmount NUMERIC(10,2),
PaymentDelay_Days INT,
OutstandingBalance NUMERIC(10,2),
FOREIGN KEY (DateKey) REFERENCES gold.DimDate(DateKey),
FOREIGN KEY (EmployeeKey) REFERENCES gold.DimEmployee(EmployeeKey),
FOREIGN KEY (CustomerKey) REFERENCES gold.DimCustomer(CustomerKey),
FOREIGN KEY (LocationKey) REFERENCES gold.DimLocation(LocationKey),
FOREIGN KEY (AccountsEmployeeKey) REFERENCES gold.DimEmployee(EmployeeKey),
FOREIGN KEY (DeliveryMethodKey) REFERENCES gold.DimDeliveryMethod(DeliveryMethodKey)
);

-- ------------------------------------------------------------------------
-- Section 5 — Indexes and partial-unique constraints
-- ------------------------------------------------------------------------

-- FactSales indexes
CREATE INDEX IF NOT EXISTS idx_factsales_datekey ON gold.factsales(datekey);
CREATE INDEX IF NOT EXISTS idx_factsales_customerkey ON gold.factsales(customerkey);
CREATE INDEX IF NOT EXISTS idx_factsales_employeekey ON gold.factsales(employeekey);
CREATE INDEX IF NOT EXISTS idx_factsales_productkey ON gold.factsales(productkey);
CREATE INDEX IF NOT EXISTS idx_factsales_locationkey ON gold.factsales(locationkey);

-- FactInvoices indexes
CREATE INDEX IF NOT EXISTS idx_factinvoices_datekey ON gold.factinvoices(datekey);
CREATE INDEX IF NOT EXISTS idx_factinvoices_customerkey ON gold.factinvoices(customerkey);
CREATE INDEX IF NOT EXISTS idx_factinvoices_employeekey ON gold.factinvoices(employeekey);
CREATE INDEX IF NOT EXISTS idx_factinvoices_accountsemployeekey ON
gold.factinvoices(accountsemployeekey);
CREATE INDEX IF NOT EXISTS idx_factinvoices_locationkey ON gold.factinvoices(locationkey);
CREATE INDEX IF NOT EXISTS idx_factinvoices_deliverymethodkey ON
gold.factinvoices(deliverymethodkey);

-- DimDate index
CREATE INDEX IF NOT EXISTS idx_dimdate_date ON gold.dimdate(date);

-- Partial-unique indexes for SCD2 current rows
CREATE UNIQUE INDEX IF NOT EXISTS ux_dimemployee_personid_current ON
gold.dimemployee(personid) WHERE date_to IS NULL;
CREATE UNIQUE INDEX IF NOT EXISTS ux_dimcustomer_customerid_current ON
gold.dimcustomer(customerid) WHERE date_to IS NULL;
CREATE UNIQUE INDEX IF NOT EXISTS ux_dimlocation_locationid_current ON
gold.dimlocation(locationid) WHERE date_to IS NULL;
CREATE UNIQUE INDEX IF NOT EXISTS ux_dimproduct_stockitemid_current ON
gold.dimproduct(stockitemid) WHERE date_to IS NULL;
CREATE UNIQUE INDEX IF NOT EXISTS ux_dimdeliverymethod_deliverymethodid_current ON
gold.dimdeliverymethod(deliverymethodid) WHERE date_to IS NULL;
```

```
CREATE UNIQUE INDEX IF NOT EXISTS ux_dimpaymentmethod_paymentmethodid_current ON
gold.dimpaymentmethod(paymentmethodid) WHERE date_to IS NULL;
```